Sri Lanka Institute of Information Technology

Artificial Intelligence and Machine Learning - IT2011

Year 2 – Semester 1

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Student ID	Name
IT24103569	Wanigasekara W.M.S.N.M
IT24103535	Sawandi T.G.A
IT24103504	Yunidu E.D.P
IT24103555	Senan R.A.D.T
IT24103517	Abesundara N. S
IT24103554	Perera K.V. N

1.Introduction and Problem Statement

Developing accurate and fair loan eligibility models helps financial institutions make faster and more consistent credit decisions. However, automated systems can unintentionally replicate or increase existing biases found in historical lending data. Therefore, fairness and ethical considerations are essential in designing such predictive models.

The goal of this project is to build a predictive model that classifies loan applicants as either **Eligible** or **Not Eligible** for a loan based on their financial and demographic information. The project also focuses on identifying and reducing potential bias against protected groups such as gender and age categories.

Objectives

- To clean and preprocess the dataset for model development.
- To train and compare multiple machine learning models, including Logistic Regression, Random Forest, XGBoost, and SVM.
- To evaluate model performance using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.
- To assess and mitigate bias using fairness measures like demographic parity and equal opportunity.
- To provide ethical and practical recommendations for responsible deployment of the model.

2. Dataset Description

The dataset contains applicant-level financial and demographic information together with the historical loan status, which serves as the target variable.

- **Structure -** Number of records (applicants): *70,000*

- **Main attributes**: loan_status (binary), person_age, person_gender, person_education, person_income, person_emp_exp, person_home_ownership, loan_intent, loan_int_rate, loan_percent_income, cb_person_cred_hist_length, credit_score, previous_loan_defaults_on_file, loan_percent_income

- **Target Variable**
    - 1 — Approved
    - 0 — Rejected

- **Sensitive Attributes** - Attributes such as gender, age_group (if available), and property_area are considered sensitive because they may relate to social and economic groups that deserve fair consideration.

- **Limitations**
  The dataset may include potential sampling bias, missing or inconsistent values, and self-reported data such as income. Additionally, the dataset might not fully represent all demographic groups, and historical loan decisions may contain label noise or bias.

3. Preprocessing & Exploratory Data Analysis (EDA)

➢ Data cleaning

- Inspect missingness patterns and record which columns contain missing values to guide imputation decisions.

➢ Outlier treatment

- Detect outliers using robust methods (IQR rule: values below Q1 − 1.5·IQR or above Q3 + 1.5·IQR) and visualize with boxplots.
- Treat extreme but plausible values by winsorization (clipping to percentile boundaries) or log-transform highly skewed variables (e.g., incomes) to stabilize variance.
- Document any records removed or heavily altered.

➢ Encoding categorical variables

- Label encoding was used for binary variables such as gender, previous_loan_defaults_on_file, and loan_status.
- One-hot encoding was applied to nominal variables like home_ownership, person_education, and loan_intent.
- E ncoding was done using a scikit-learn pipeline to prevent data leakage.

➢ Scaling

- Apply StandardScaler (zero mean, unit variance) to numeric features used by distance-based or linear models (Logistic Regression, SVM).
- Avoid scaling for tree-based models (Random Forest) as they are scale-invariant.
- Ensure scaling is performed inside the pipeline after imputation and before model fitting to prevent data leakage.

➢ Feature selection

- Conduct univariate correlation analysis and remove features with very high multicollinearity.

- Use model-based importance (e.g., Random Forest feature importance) and wrapper methods (recursive feature elimination) to identify and retain the most predictive features.
- Validate selected features by comparing cross-validated performance with the full feature set and by checking stability of selected features across folds.

EDA — key checks and visualizations

- Plot distributions for numeric features (histograms, log-transformed histograms) and categorical counts.
- Show approval rates by sensitive attributes (e.g., gender, property_area) to surface any disparities.
- Include a correlation heatmap, pairwise scatterplots for top features, and ROC/precision-recall curves for baseline models.
- Summarize findings from EDA that inform modeling choices (e.g., skewed income distribution, missing credit history concentration in specific groups).

4. Model Design and Implementation

We trained and compared these classifiers:

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbors (KNN)
- Random Forest
- Multi-Layer Perceptron (MLP) Neural Network

Training procedure

- All preprocessing and modeling were wrapped in scikit-learn pipelines (imputation → encoding → scaling → estimator) to prevent leakage.
- Scaling applied to scale-sensitive models; tree-based models received raw features.
- Handled class imbalance via class weights or training-fold-only resampling (SMOTE).
- Hyperparameter tuning via stratified 5-fold cross-validation using randomized or grid search. Optimization objective chosen based on business priorities (F1 or ROC-AUC).
- Feature selection validated through cross-validation (recursive feature elimination and permutation importance).

Interpretability

- Logistic Regression coefficients and tree-based feature importances provide global insights.
- SHAP values used for local and global explanations of complex models (Random Forest, MLP).
- Model cards were created for the chosen model documenting inputs, performance, limitations, and recommended human review processes.

5. Evaluation and Comparison

➢ Evaluation procedure

- Stratified 80/20 train/test split; hyperparameter tuning on training set with stratified 5-fold CV.
- Resampling applied only within training folds to avoid leakage.
- Final performance measured on held-out test set.

➢ Performance metrics

Reported metrics for each model include:

- Accuracy
- Precision
- Recall (Sensitivity)
- F1-score

Results summary

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.879214	0.912996	0.838963	0.874415
SVM	0.887929	0.908396	0.863474	0.885366
Decision Tree	0.895071	0.875169	0.922189	0.898064
KNN	0.875286	0.887973	0.859627	0.873570
Random Forest	0.910500	0.915633	0.904803	0.910186
MLP Nural Network	0.889071	0.903664	0.871598	0.887341

Comparison:

Based on the performance comparison of different models, the Random Forest algorithm achieved the highest overall performance with an accuracy of 0.9105, precision of 0.9156, recall of 0.9048, and an F1-score of 0.9102. The Decision Tree model also performed well, showing a relatively high recall (0.9222) but slightly lower precision. Other models such as SVM and MLP Neural Network showed balanced performance, while Logistic Regression and KNN achieved comparatively lower scores.

- Therefore, the **<u>Random Forest</u>** model was selected as the best-performing model for this study.

6. Ethical Considerations and Bias Mitigation

- **Privacy and consent:** Ensure all data collection complied with informed consent; anonymize identifiable attributes.
- **Fairness:** Do not remove sensitive attributes blindly; instead measure and mitigate bias using principled methods and stakeholder input.
- **Transparency:** Publish a model card describing data sources, metrics, limitations, and intended use.
- **Human oversight:** Incorporate human review for borderline or high-impact decisions and maintain an appeal or feedback mechanism for applicants.
- **Monitoring:** Continuously monitor model performance and fairness metrics after deployment and retrain when dataset shift is detected.

7. Reflections and Lessons Learned

- Data quality and representativeness are critical; missing or biased labels can mislead models.
- Preprocessing choices (imputation, encoding, scaling) materially affect model performance and fairness.
- Fairness interventions require trade-offs; engage stakeholders to set acceptable thresholds.
- Reproducible pipelines and saved artifacts (preprocessors, model objects) are essential for auditability and deployment.

8. References

- Dataset: https://www.kaggle.com/datasets/taweilo/loan-approval-classification-data

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